Can an LLM Estimate the Embodied Carbon of a Building Enclosure?

Error Analysis, Prompt Design, and Implications for LEED v5 Embodied Carbon Credits

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Abstract

We test whether a large language model (Claude Opus 5.5, run as a Claude Code subagent with medium reasoning effort) can estimate the cradle-to-gate (A1–A3) embodied carbon of the building enclosure from early-stage design features. Reference values come from the 292 buildings of the harmonized North American WBLCA dataset of Benke et al. [1].

With a plain prompt, only 19.5% of estimates fall within ±20% of the reference LCA. Error analysis shows that the model compresses all buildings toward a typical value and never considers HFC-blown XPS insulation, which is 15.5% of enclosure carbon and about half of the systematic underestimate. It also underestimates envelope area and misjudges renovations; its largest overestimates trace back to incomplete reference LCAs.

Adding qualitative knowledge and a step-by-step method made the model more consistent but less accurate. Placing externally sourced quantitative anchors at the step where each number is chosen raised coverage to 25.0%, moved the bias (geometric mean of estimate/reference) from 0.75 to 0.86, and cut run-to-run variation by a third. On 88 held-out buildings the gains point the same way but are not significant. Placing envelope-focused documents in context instead made the estimates worse (bias 0.61): the documents describe current low-carbon products rather than the data the reference LCAs used. Ten early-design inputs are as informative as all 48, and building use shapes enclosure carbon more than climate.

For LEED v5, the answers name the reference's top hotspot 95% of the time and support screening and the comparison of envelope alternatives at schematic design (MRp2, MRc2). They cannot document a credit-level reduction for an individual building.

1 Introduction

Early design decisions fix most of a building's embodied carbon. The exterior enclosure is among the most consequential and variable of them. Early-stage LCA is hard because design information is sparse and the hand-off to an LCA practitioner is slow. LLM in-context learning has been shown to support early-stage whole-building LCA, with domain knowledge improving on innate knowledge [2].

We narrow that question to the enclosure and to the product stage (A1–A3), where material choices dominate and where LEED v5 embodied-carbon requirements start. We ask:

Q1 How accurately and consistently does an LLM estimate enclosure A1–A3 carbon from early-stage features, and why does it fail?

Q2 Which prompt changes help, and which do not?

Q3 Which input features does the model need?

Q4 What do the answers imply for the LEED v5 embodied-carbon prerequisite and credit?

Scope relative to the proposal. The proposal planned a reference LCA of three modelled wall assemblies in two to three climates and two uses, per m^2 of wall and in six TRACI categories. It also planned three knowledge conditions with 10 runs each, and an evaluation of LEED credit outcomes. After the course check-in we adopted the course's dataset-based design, which changed four things:

- *Ground truth.* Reported enclosure A1–A3 GWP of 292 real projects replaces modelled assemblies. Only GWP is evaluated, and the unit is per m^2 of floor area.
- *Climate and use (proposal RQ1).* These are analysed from the data (Section 2), not from modelled assemblies.
- *Knowledge conditions.* We test the innate baseline, a domain-knowledge condition with envelope-focused documents (Section 6) and two prompt revisions. The ontology condition was not run. To limit cost we used 3 runs (1 for the domain and feature conditions) instead of 10.
- *LEED.* MRp2 hotspot agreement is measured with a mention-based proxy (Section 9). MRc2 point-band agreement, the LEED-aligned ontology and the cost/regulation/performance drivers are discussed only qualitatively or left out.

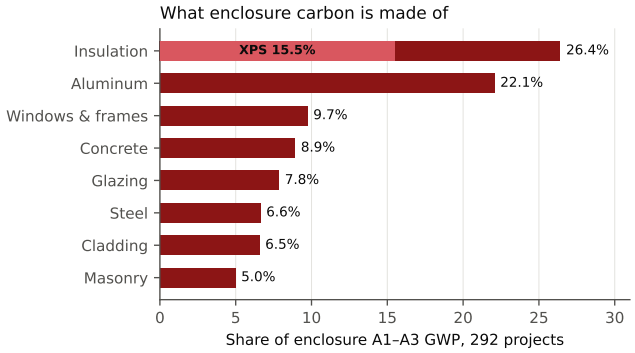


Figure 1: Share of enclosure A1–A3 GWP by material group, 292 projects. XPS alone is 15.5%.

2 Data and Ground Truth

Source. The harmonized dataset of Benke et al. [1] (CC BY 4.0) holds 927,950 material-level LCA results for 292 projects in the US and Canada, with 48 early-stage features per project.

Scope and unit. We keep OmniClass `Shell - Enclosure` and stage `A1-A3` (55,807 rows) and divide by floor area. We use `bldg_cfa` for new construction and renovated plus added GFA for the 49 renovations, which reproduces the dataset's own normalization for all 292 projects. The enclosure includes windows, doors and roofs, and the unit is per m^2 of floor area rather than the proposal's 1 m^2 of wall, because wall area is missing for most projects.

Distribution. Values have a median of 78.0 $kgCO_2e/m^2$, a mean of 95.6 and a range of 0.49–513.9. The enclosure is a median 19% of whole-building A–C carbon.

Climate and use (proposal RQ1). We looked at the 233 new-construction projects with complete references (231 with a climate zone). Building use matters (Kruskal–Wallis $p < 10^{-6}$): medians run from 53.6 $kgCO_2e/m^2$ for multifamily and 64.5 for warehouses, through 87.2 for offices and 89.3 for education, to 132.6 for public safety and 157.2 for transport hubs. Climate does not reach significance ($p = 0.26$): the medians are 80.6 in warm zones (1–3), 85.9 in zone 4 and 95.1 in cold zones (5–8). Per m^2 of floor area, use (through glazing share and envelope-to-floor ratio) shapes enclosure carbon more than climate (through insulation).

Composition. Insulation (26.4%) and aluminum (22.0%) dominate (Fig. 1). XPS alone is 15.5%, at 43.6 $kgCO_2e/kg$, against 1.2 for mineral wool and 3.1 for polyiso. That is the HFC-blown product modelled by Tally and One Click LCA in the assessment years. 149 projects contain XPS.

Data quality. Ten new-construction projects have no fenestration GWP at all, although several report a WWR of 0.5–0.7. We report results with and without them.

3 Methods

Model and isolation. Each estimate comes from a Claude Code subagent (Claude Opus 5.5, medium effort). The subagent may read only its task file and write only its answers; it has no web access, no code execution and no access to the ground truth. A task file holds the system prompt, the feature glossary and 48–49 building blocks, each with the building's features and the task text (Appendix A). The model is told to treat every building independently.

Protocol. Each prompt version is run 3 times on all 292 buildings (876 estimates). Buildings are shuffled with a fixed seed per run and split into six chunks. The chunks are identical across prompt versions, so results can be paired building by building. The estimate is the mean of the midpoints of the returned ranges. In one v3 run a container restart interrupted the subagents; the missing 162 buildings were answered by continuation tasks with the same prompt.

Parsing. We use the course parser (the last line `RANGE_kgCO2e_per_m2: low - high`) but accept a lower bound of 0. Renovations that leave the envelope untouched have reference values near zero, and the original parser would have discarded exactly those answers.

Metrics.

- Coverage: the share of buildings whose reference lies within the estimate $\pm 20\%$ (Wilson CI).
- RMSE, median absolute percentage error (APE), and the median $|\log(\text{est}/\text{ref})|$.
- Bias: the geometric mean of est/ref.
- Consistency: the coefficient of variation (CV) across runs.
- Paired tests between prompts: McNemar on coverage, Wilcoxon signed-rank on $|\log|$ error.

A test set of 88 buildings is held out, stratified on project type and climate group (seed 226).

4 Baseline and Error Analysis

Baseline results. With prompt v1, coverage is 19.5% (CI 15.4–24.5%), RMSE 79.7 $kgCO_2e/m^2$, median APE 37.9%, bias 0.75 and CV 19.0% (Table 1). 63% of buildings are underestimated beyond the band, and 17.5% overestimated.

Compression. The log–log slope of estimate on reference is 0.46. Buildings below 25 $kgCO_2e/m^2$ are overestimated 1.58×, and buildings above 200 are estimated at 0.28× (Fig. 2). The 20 largest errors, all underestimates, carry 68% of the squared error.

Geometry over material. The model leans on geometry: the envelope-to-floor ratio correlates 0.62 with the log estimate but only 0.39 with the log reference. What it misses is the carbon intensity per m^2 of envelope, which correlates −0.79 with the log error. That intensity is set by assembly and material choices, and *no feature describes the wall assembly*.

Why the model underestimates.

1. *HFC-blown XPS is never considered.* Only 0.6% of answers mention XPS, and none mention blowing agents. Projects with XPS sit at 0.67, projects without at 0.80. Removing XPS from the reference raises the median ratio from 0.72 to 0.84, so XPS explains about half of the bias.
2. *The climate gap is an insulation effect.* Cold zones sit at 0.65 and warm zones at 0.77. XPS occurs in 65–68% of zone 5A/5B projects but only 26–33% of zone 3B/3C projects.
3. *Missing envelope area.* The ratio is 0.81 when the envelope area is given and 0.65 when the model must estimate it.
4. *Curtain-wall tail and renovations.* Single projects reach 300 $kgCO_2e/m^2$ from curtain wall alone. 60% of major renovations are underestimated.
5. *Scope wording.* The v1 scope was narrower than the dataset's enclosure; for example, 11% of enclosure GWP was tagged "Superstructure" in the source tool.

Why the model overestimates. The ten incomplete references have a median ratio of 3.09. The other overestimates are mostly fit-outs and minor renovations whose extent of envelope work is unknown.

5 Prompt Iterations

5.1 Prompt v2: qualitative knowledge and method

Prompt v2 changed six things:

- the scope was written to match the dataset;
- a component take-off was required (areas → assemblies → intensities → total);
- high-impact materials were named: HFC-blown XPS "ten times or more" the GWP of other insulation per unit of thermal resistance [4, 5], and aluminum curtain wall at 100–200 $kgCO_2e/m^2$ [6, 7];
- anchoring on a typical building was ruled out;
- renovations got their own rule;
- a `BEST` value was requested alongside the range.

It made the model more consistent (CV 13.8%) but *less* accurate: coverage 15.1% and bias 0.67, with significantly larger errors (Wilcoxon $p < 0.0001$). Three reasons:

1. *The facts were repeated, not applied.* XPS was named 50× more often, but 89% of answers still assumed polyiso roof insulation, and the median roof intensity stayed at 35 $kgCO_2e/m^2$.
2. *The take-offs were too light.* Bottom-up intensities came out at 66.8 per m^2 of envelope against 86.7 in the reference.
3. *A subtracting rule hurt.* "Count only new materials" dropped minor renovations from 0.85 to 0.56.

5.2 Prompt v3: quantitative anchors

v3 keeps the v2 scope and method. A separate research subagent compiled a reference table of A1–A3 intensities per m^2 of component from external sources [3, 9, 8, 10], excluding the ground-truth dataset. v3 inserts this table at the step where numbers are chosen, and adds:

- the modelling convention that LCA tools of the period often used HFC-blown XPS for roofs;
- a check against roughly 50 $kgCO_2e/m^2$ per m^2 of envelope;
- a renovation rule that no longer subtracts.

The full text is in Appendix A.6.

Results. v3 is best on coverage (25.0%), RMSE (76.0), $|\log|$ error (0.426), bias (0.86) and CV (12.5%) (Table 1, Fig. 3).

- The gain over v2 is significant ($p = 0.002$). The gain over v1 is consistent but borderline (McNemar $p = 0.085$, Wilcoxon $p = 0.053$).
- HFC-XPS is now considered in 96% of answers for projects that contain XPS.
- The non-XPS part of the enclosure (0.99), buildings with a given envelope area (0.99) and warm zones (0.98) are now unbiased.
- Remaining problems: compression at the top (0.38 above 200 $kgCO_2e/m^2$), more overestimation of light enclosures (1.31 in the 25–50 bin), and renovations.

6 Domain Knowledge from Envelope Documents

The course design tests domain knowledge by placing reference documents in the context. Following the check-in, we replaced the general whole-building papers with four envelope-focused documents, each truncated to 30k characters as in the course pipeline:

- RDH Building Science's wall-assembly LCA (2025) [10];
- the CLF 2023 North American Material Baselines [3];
- the Arup/Saint-Gobain study of facades and glass [7];
- a UBC study of fenestration embodied carbon [11].

The task text is v1 unchanged, so the documents are the only difference from the baseline. The condition was run once and compared building by building with run 0 of v1 and v3, which use the same chunks and order (Table 2).

Results. The documents made the estimates *worse*:

- The median $|\log|$ error rose by 0.090 (Wilcoxon $p < 0.001$), the bias moved from 0.72 to 0.61, and only 35% of buildings got closer. The drop in coverage (-2.1 points) is not significant ($p = 0.48$).
- v3 beats the document condition on every metric (coverage $+8.6$ points, McNemar $p = 0.011$; $|\log|$ error -0.148, $p < 0.001$).

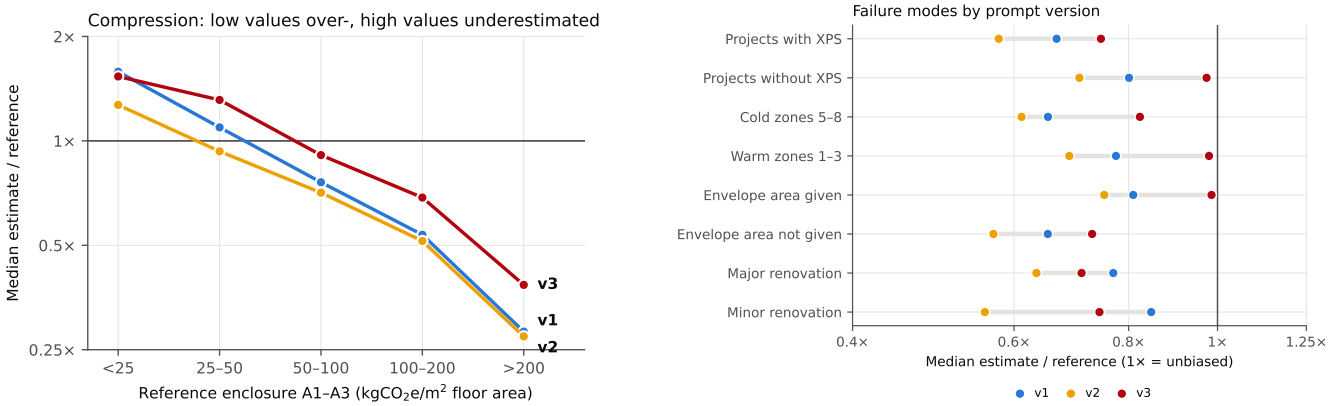


Figure 2: Left: median estimate/reference by reference bin. All prompts compress; v3 shifts the curve up. Right: subgroups behind each failure mode, by prompt.

Table 1: Accuracy and consistency by prompt (3 runs each). Coverage: reference within estimate ±20%. Bias: geometric mean of estimate/reference.

Buildings	Prompt	Coverage (95% CI)	RMSE	Median APE	Median $\lvert\log\rvert$	Bias	CV
All (292)	v1	19.5% (15.4–24.5)	79.7	**37.9%**	0.456	0.75	19.0%
	v2	15.1% (11.4–19.6)	81.0	44.3%	0.563	0.67	13.8%
	v3	**25.0%** (20.4–30.3)	**76.0**	40.0%	**0.426**	**0.86**	**12.5%**
Held-out test (88)	v1	20.5% (13.3–30.0)	74.1	**36.4%**	**0.393**	0.75	19.2%
	v2	18.2% (11.5–27.5)	75.7	43.4%	0.511	0.66	15.2%
	v3	**26.1%** (18.1–36.2)	**69.2**	37.0%	0.400	**0.86**	**11.2%**

Table 2: Envelope documents in context (run 0, 292 buildings). p: Wilcoxon on $\lvert\log\rvert$ error, paired against the row above.

Condition	Cov.	Med. $\lvert\log\rvert$	Bias	RMSE	p
v1 (innate)	17.8%	0.519	0.72	81.2	–
v1 + documents	15.8%	0.609	0.61	85.3	< 0.001
v3 (final)	24.3%	0.461	0.86	77.8	< 0.001

Why. The documents were read: 22% of answers cite them, mostly to justify the low end of a range ("per RDH", "CLF baselines support the lower end"). But they describe *current* lower-carbon products: HFO-blown spray foam, recent EPD baselines and modern glazing. The reference LCAs were modelled with the tool data of their assessment years.

- XPS mentions fell from 0.7% of answers to zero.
- The ratio for projects with XPS fell from 0.64 to 0.50.
- The ratio without XPS also fell, from 0.78 to 0.68.

Like v2, this shows that more domain text does not correct a blind spot; it can deepen it when the text describes a different product vintage than the reference models. What worked in v3 was a short table of sourced intensities together with an explicit note on the modelling convention.

7 Held-out Evaluation

No external test set. No independent dataset with the same structure exists. The CLF WBLCA benchmark probably shares source models with the ground truth. CarbEnMats [12] has envelope A1–A3 values for only 172 European buildings, defines the element differently, and has no geometry fields.

Dev-only replication. The split was assigned after the runs. To check that the prompt design did not depend on the test buildings, we re-ran every diagnostic on the 204 dev buildings alone. Every finding behind the prompt revisions replicates:

- compression slope 0.44;
- XPS share 14.7%, with XPS effect 0.66 vs. 0.77;
- envelope-area effect 0.79 vs. 0.65;
- v2's polyiso assumption 89%.

Test results. On the 88 test buildings, v3 improves on v1 in coverage (26.1% vs. 20.5%), RMSE (69.2 vs. 74.1), bias and CV. Its median $\lvert\log\rvert$ error is unchanged (0.400 vs. 0.393), and no v1–v3 difference is significant at this size ($p = 0.41$).

8 Feature Study

With v3 fixed, each feature set was run once and paired with run 0 of v3 on all 48 features (Table 3). The sets were:

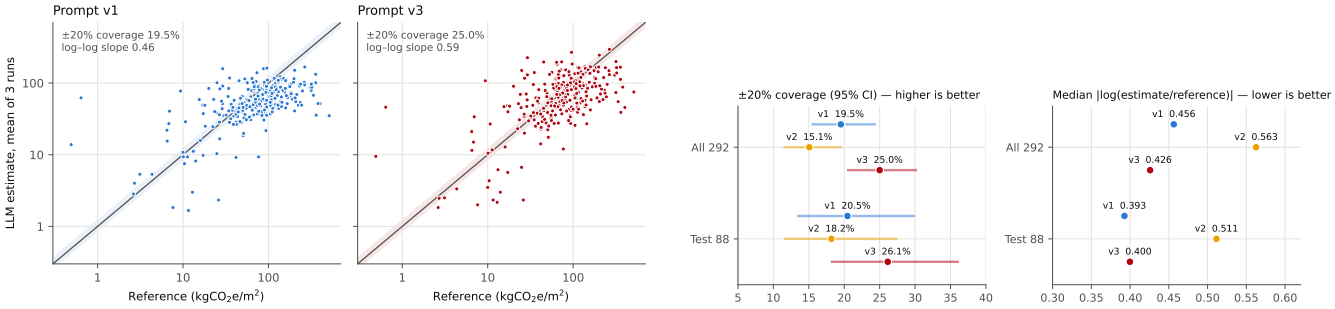


Figure 3: Left: estimate vs. reference for v1 and v3 (1:1 line, ±20% band). Right: coverage (95% CI) and median $|\log(\text{est}/\text{ref})|$ for all 292 and the 88 held-out buildings.

Table 3: Feature study (v3, run 0, 292 buildings). Δ and p (Wilcoxon) are paired against all 48 features.

Set	n_f	Cov.	Med. $\lvert\log\rvert$	Δ	p	Bias
all	48	24.3%	0.461	–	–	0.86
early	39	21.6%	0.462	+0.001	0.13	0.88
env	15	19.9%	0.463	+0.001	0.33	0.93
min	10	25.3%	0.476	+0.015	0.81	0.81
no-lca	40	25.3%	0.441	−0.020	0.57	0.90
no-mass	47	22.6%	0.452	−0.009	0.92	0.85

- *early* (39): no LCA-derived fields and no `mass_total`, the reference LCA's own inventory mass;
- *env* (15): envelope-relevant features;
- *min* (10): climate, use, type, area, storeys, WWR and R-values;
- *no-lca* (40) and *no-mass* (47): one group removed each.

Results. No set differs significantly from the full set, and every difference lies within v3's own run-to-run spread (0.025 in $|\log|$). Ten early-design inputs are as good as 48, and the model does not depend on `mass_total`. Prompt content moved the $|\log|$ error by 0.137 from v2 to v3; no feature set moved it by more than 0.020.

9 Implications for LEED v5

LEED v5 makes embodied carbon part of the rating system. Prerequisite MRp2 (*Quantify and Assess Embodied Carbon*) asks projects to quantify the embodied carbon of their main building elements, including the enclosure, and to identify the largest hotspots. Credit MRc2 (*Reduce Embodied Carbon*) rewards reductions: either a whole-building LCA compared with a baseline (Option 1), or product-level reductions documented with EPDs (Option 2). Fig. 4 shows where the enclosure LCA and the LLM estimate sit in that workflow, from project inputs and the A1–A3 system boundary to a recommended envelope design with its estimated LEED contribution and remaining data gaps.

MRp2: hotspot screening. At schematic design, no material inventory exists yet, and the prerequisite's purpose is to direct attention to the largest sources. The v3 estimates suit that purpose.

- The bias is small, −14% overall and close to zero for buildings without XPS.
- The reasoning names the components that drive enclosure carbon in the data: aluminum curtain wall and framing, insulation type (HFC-XPS), metal panels and glazing.

We tested hotspot agreement with a proxy. Each building's top-3 hotspots are the three largest of 11 design-level material categories in the reference; an answer counts as naming a category when it mentions the category (keyword match, `analysis/hotspot_agreement.py`). Excluding the 10 incomplete references:

- v3 answers name the reference top-1 hotspot in 94.6% of cases and recall 89.7% of the top-3. Naming the same number of categories at random would give 60.5%; v1 reaches 81.4% and 77.1%.
- The answers are building-specific. For the 80 buildings whose top-1 hotspot is not one of the three most common categories (insulation, glass, aluminum), v3 names the top-1 in 82.1% of cases, where a constant typical-building guess scores 0%.
- The answers are not a ranked shortlist: they name 6.7 categories on average, and their precision (41%) is below that of the typical-building guess (60%). For MRp2 the prompt should ask for a ranked top-3 explicitly.

MRc2: reductions need precision the LLM does not have. Credit points scale with the percentage reduction against a baseline. The relevant errors are:

- the run-to-run CV of v3 (12.5%);
- its median absolute error (about 35–40%);
- coverage of only one building in four within ±20%.

These are as large as, or larger than, the reductions a project typically documents, so an absolute LLM estimate cannot demonstrate a credit-level reduction for an individual building. The more promising use is a *paired* comparison of envelope alternatives for the same building with the same prompt,

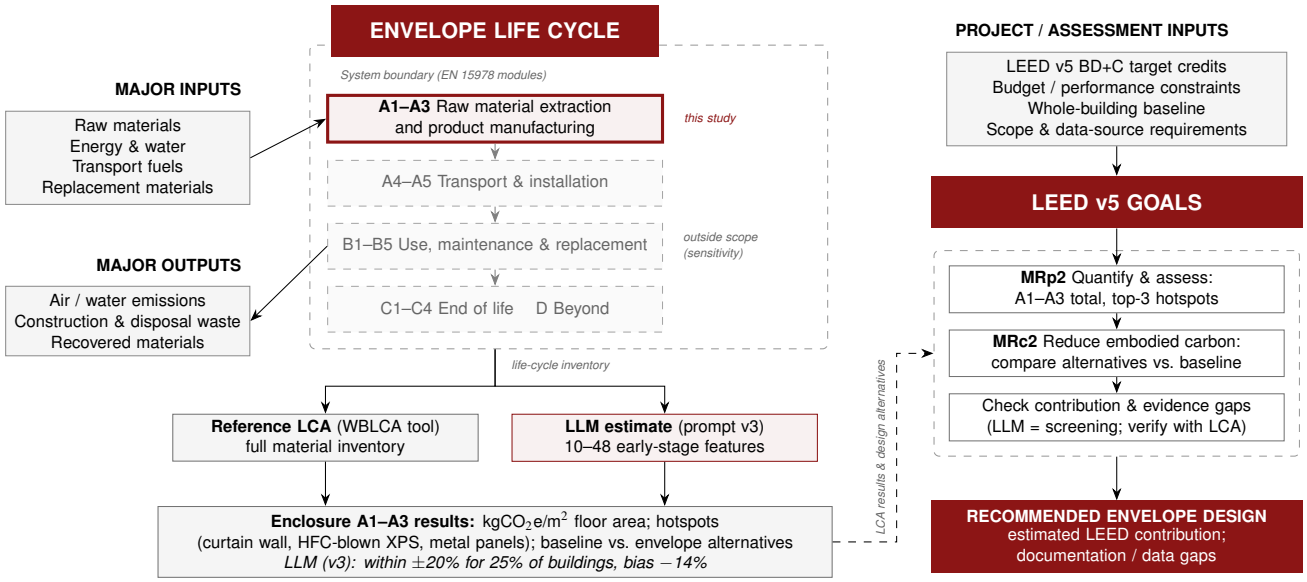


Figure 4: Building-envelope LCA and LEED v5 goals. The study boundary is the product stage (A1–A3); later stages are outside scope. A reference LCA and the LLM estimate (prompt v3) both produce enclosure A1–A3 results. Those results feed the LEED v5 workflow: MRp2 (quantify and assess embodied carbon, top hotspots) and MRc2 (reduce embodied carbon against a baseline). The LLM serves for screening and comparing alternatives, and its results are verified by an LCA before documentation.

where shared biases partly cancel. That comparison was not tested here and needs an LCA for documentation.

Option 1: enclosure gains are diluted. In this dataset the enclosure is a median 19% of whole-building A–C carbon. A 30% enclosure reduction is therefore only about 6% at the whole-building level. Envelope decisions alone rarely carry a whole-building reduction target; they are most valuable in combination with structure and in Option 2's product-level pathway. There, high-impact products such as HFC-blown XPS, aluminum extrusions and insulated metal panels have the largest EPD-to-EPD spreads (Appendix A.6).

Making early estimates more predictable. Three changes would make early estimates and LEED baselines more predictable:

1. report assembly type at schematic design;
2. state which insulation and aluminum datasets the LCA tool assumes, since the HFC-XPS convention alone moved our estimates by about half of the bias;
3. flag inventories with missing element groups. The 10 incomplete references in this dataset would distort any baseline built from it.

10 Limitations

- Answers in one task file share a context, so they are less independent than separate API calls.
- The held-out split was assigned after the runs.
- The v3–v1 gain is borderline with 3 runs.
- The document condition used one run and four documents truncated to 30k characters; other documents could behave differently.
- The feature study used one run per condition, and 7 of 9 planned ablations were stopped to limit cost.
- The unit is per m^2 of floor area.
- Reference quality is uneven.
- Hotspot agreement uses a mention-based proxy, and MRc2 point-band agreement was not computed.
- Results are specific to Claude Opus 5.5.

11 Conclusion

A frontier LLM can produce enclosure A1–A3 estimates from early-stage features. Its error is structured: compression toward a typical value, and blind spots for specific high-carbon materials. Error analysis of its reasoning located those blind spots. Sourced quantitative anchors placed at the decision step moved the bias from 0.75 to 0.86, cut variation by a third, and raised coverage to 25%. Qualitative instructions (v2) and envelope documents in context both made the estimates worse.

For LEED v5, this supports early hotspot screening and the comparison of alternatives, not credit documentation. Further progress depends on giving the model the assembly information that early-stage feature sets lack.

References

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[12] M. Röck et al. CarbEnMats-Buildings: a global database on whole life carbon, energy and material intensity of buildings. Zenodo, 2023.

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A Prompts

Exported verbatim from `experiment_config.py` by `report/latex/make_prompts.py`. Six non-ASCII glossary characters are shown as ASCII (e.g. CO2). Each request consists of the system prompt, the shared context, and a building block (features followed by the task text).

A.1 System prompt

```
You are a building LCA expert.
```

A.2 Shared context (identical in v1, v2 and v3)

```
Forget the information from previous sessions and calculate everything from the beginning.

data_glossary_LLM contains feature names and descriptions for the data:

- project_index: Unique project index number
- site_country: Country where the project is located
- site_region: Region where the project is located. Based on the 2019 National Ready Mix Concrete Association regions.
- site_state_province: State or province where the project is located.
- site_clim_zone: Climate Zone where the project is located per respective energy code of country.
- bldg_proj_type: Type of construction which details whether the project is new or a type of modification to an existing building.
- bldg_compl_year: Year of project's actual construction completion, certificate of occupancy, or anticipated project completion
    year.
- bldg_code_year: Corresponding year of the international building code (IBC) or other equivalent the project complied with.
- bldg_energy_code_year: Corresponding year of the international energy conservation code (IECC) or other equivalent the project
    complied with.
- bldg_ibc_type: Type of construction per IBC.
- bldg_park_type: Type of attached/integrated parking structure or "parkade".
- bldg_cfa [m2]: Total constructed floor area (CFA) which includes the gross internal floor area (GFA) of the building and the gross
    floor area of any attached/integrated parking components.
- bldg_gfa [m2]: Total gross internal floor area (GFA) of the building.
- bldg_parking_gfa [m2]: The total horizontal area occupied by any attached/integrated parking components.
- bldg_added_gfa [m2]: The total horizontal area where new floor area was added to an existing building (i.e., the floor area of an
    addition).
- bldg_renovated_gfa [m2]: The total horizontal area where the existing building was renovated.
- bldg_prim_use: Primary classification of the building by principal activity
- bldg_sec_use: Secondary classification of building by principal activity.
- bldg_occupants: Total building occupancy as derived from building code
- bldg_res_units: Total number of residential units in the building
- bldg_stories_above: Binned categorization of the number of stories above finished ground level.
- bldg_stories_below: Binned categorization of the number of stories below finished ground level.
- bldg_height [m]: Binned categorization of the total height of the building above finished ground level.
- bldg_therm_env_area [m2]: Total area of the building's thermal envelope.
- bldg_wwr: The ratio of total vertical fenestration area to total above-grade wall area
- bldg_rval_walls: Average nominal R-value of building's above-grade walls.
- bldg_rval_roofs: Average nominal R-value of building's roofs.
- str_seis_site_cls: Classification of the site for seismic design based on its soil and engineering properties.
- str_sdc: Categorization of required seismic design based on risk category and site location.
- str_wind_speed [m/s]: Ultimate wind speed used for ultimate wind design based on location.
- str_prim_horiz_sys: Type of primary horizontal gravity system. Structural floor construction system that supports at least 2/3 of
    the combined floor and roof area of the superstructure.
- str_prim_vert_sys: Type of primary vertical gravity system. System that transfers gravity loads to the foundation of the building.
- str_lat_sys: Type of lateral system that resists lateral loads of the building.
- str_podium: Designation for whether the project is a podium building - a building with two distinct zones with two different
    structural materials and systems
- str_sec_horiz_sys: The horizontal gravity system (as defined in Primary Horizontal Gravity System) for the podium portion of the
    building.
- str_sec_vert_sys: The vertical gravity system (as defined in Primary Vertical Gravity System) for the podium portion of the
    building
- str_grid_long [m]: Average center to center spacing of longer typical project column grid.
- str_grid_short [m]: Average center to center spacing of shorter typical project column grid.
- str_fdn_type: Type of foundation used to support gravity and lateral loads of project.
- str_sys_summary: Simplified summary of the buildings primary horizontal, vertical, and lateral system.
- lca_assessment_year: Year the LCA was performed
- lca_design_phase: The design phase of the project when the LCA was performed
- lca_RSP [years]: Reference study period for the LCA
- lca_software: Software tool used for the LCA
- lca_purp_of_assessment: Primary reason the LCA was undertaken.
- lca_phys_scope: The physical scope (building elements) included in the assessment.
- lca_ec_reductions: Yes/No designation to indicate if the WBLCA results reflect efforts by the design or construction team to
    reduce the embodied carbon of the project during design, procurement, and/or construction of the building.
- lca_ec_reduction_percent: Approximate percent reduction in embodied carbon of the building compared to a baseline model, if
    applicable.
- mass_total [kg]: Total mass of project. Sum of the project's total mass (inv_mass) for life cycle stages A-C
- stored_carbon_total [kg CO2e]: Total stored carbon of project. Sum of the project's total stored carbon (inv_stored_carbon) for
    life cycle stages A1-A3
```

```
- gwp_a_to_c [kg CO2e]: Sum of the project's total global warming potential (gwp) for life cycle stages A-C
- ap_a_to_c [kg SO2e]: Sum of the project's total acidification potential (ap) for life cycle stages A-C
- ep_a_to_c [kg Ne]: Sum of the project's total eutrophication potential (ep) for life cycle stages A-C
- sfp_a_to_c [kg O3e]: Sum of the project's total smog formation potential (sfp) for life cycle stages A-C
- odp_a_to_c [kg CFC-11e]: Sum of the project's total ozone depletion potential (odp) for life cycle stages A-C
- nred_a_to_c [MJ]: Sum of the project's total non-renewable energy demand (nred) for life cycle stages A-C
- mui_total_GFA [kg/m3]: Total material use intensity. Sum of the project's total material use intensity (mui_gfa) over the full RSP
      normailized by GFA for new construction projects. Includes material replacements from module B.
- eci_a_to_c_GFA [kgCO2e/m2]: Embodied carbon intensity. Sum of the project's total global warming potential (gwp) for life cycle
      stages A-C normalized by GFA for new construction projects.
- epi_a_to_c_GFA [kgNe/m2]: Eutrophication potential intensity. Sum of the project's total eutrophication potential (ep) for life
      cycle stages A-C normalized by GFA for new construction projects.
- api_a_to_c_GFA [kgSO2e/m2]: Acidification potential intensity. Sum of the project's total acidification potential (ap) for life
      cycle stages A-C normalized by GFA for new construction projects.
- sfpi_a_to_c_GFA [kgO3e/m2]: Smog formation potential intensity. Sum of the project's total smog formation potential (sfp) for life
      cycle stages A-C normalized by GFA for new construction projects.
- odpi_a_to_c_GFA [kgCFC11e/m2]: Ozone depletion potential intensity. Sum of the project's total ozone depletion potential (odp) for
      life cycle stages A-C normalized by GFA for new construction projects.
- nredi_a_to_c_GFA [MJ/m2]: Non-renewable energy demand intensity. Sum of the project's total non-renewable energy depletion
      intensity (nredi) for life cycle stages A-C normalized by GFA for new construction projects.
- mui_total_CFA [kg/m2]: Total material use intensity. Sum of the project's total material use intensity (mui_cfa) over the full RSP
      normailized by CFA for new construction projects. Includes material replacements from module B.
- eci_a_to_c_CFA [kgCO2e/m2]: Embodied carbon intensity. Sum of the project's total global warming potential (gwp) for life cycle
      stages A-C normalized by CFA for new construction projects.
- epi_a_to_c_CFA [kgNe/m2]: Eutrophication potential intensity. Sum of the project's total eutrophication potential (ep) for life
      cycle stages A-C normalized by CFA for new construction projects.
- api_a_to_c_CFA [kgSO2e/m2]: Acidification potential intensity. Sum of the project's total acidification potential (ap) for life
      cycle stages A-C normalized by CFA for new construction projects.
- sfpi_a_to_c_CFA [kgO3e/m2]: Smog formation potential intensity. Sum of the project's total smog formation potential (sfp) for life
      cycle stages A-C normalized by CFA for new construction projects.
- odpi_a_to_c_CFA [kgCFC11e/m2]: Ozone depletion potential intensity. Sum of the project's total ozone depletion potential (odp) for
      life cycle stages A-C normalized by CFA for new construction projects.
- nredi_a_to_c_CFA [MJ/m2]: Non-renewable energy demand intensity. Sum of the project's total non-renewable energy depletion
      intensity (nredi) for life cycle stages A-C normalized by CFA for new construction projects.
- ec_per_occupant_a_to_c [
kgCO2e/occupant

]: Embodied carbon per occupant. Sum of the project's total global warming potential (gwp) for life cycle stages A-C normalized by
      the number of buildings occupants (bldg_occupants).
- ec_per_res_unit_a_to_c [kgCO2e/ unit]: Embodied carbon per residential unit. Summary of the project's total GWP (gwp) for life
      cycle stages A-C normalized by number of residential units (bldg_res_units).
- nan: nan
- nan: nan
- nan: nan
- nan: nan
- nan: nan
- nan: nan
- nan: nan
- nan: nan
- nan: nan
- nan: nan
- nan [kg]: nan
- nan [kg CO2e]: nan
- nan [kg CO2e]: nan
- nan [kg SO2e]: nan
- nan [kg Ne]: nan
- nan [kg O3e]: nan
- nan [kg CFC-11e]: nan
- nan [MJ]: nan
- nan [kg/m2]: nan
- nan [kg/m2]: nan

Reference the given feature information above. Estimate the embodied carbon of the building enclosure of the building below:
```

A.3 Example building features (project 225, all 48 features)

```
site_country: United States
site_region: Eastern
site_clim_zone: Zone 5A (Cool Humid)
bldg_code_year: 2015
bldg_energy_code_year: 2009
bldg_proj_type: New Construction
bldg_ibc_type: Type I-A
bldg_park_type: No Parking
bldg_cfa: 25000
bldg_gfa: 25000
bldg_prim_use: Healthcare
bldg_occupants: 2062
bldg_stories_above: 6 to 10
```

```
bldg_stories_below: 2
bldg_height: 22.6-30 m
bldg_therm_env_area: 12133.1318
bldg_wwr: 0.4
str_seis_site_cls: D
str_sdc: C
str_wind_speed: 56.77408
str_prim_horiz_sys: Steel: Frame + concrete on metal deck
str_prim_vert_sys: Steel: Columns
str_lat_sys: Steel: Moment frames
str_podium: Primary system defined above is on a podium
str_sec_horiz_sys: Concrete: Non-PT framing
str_sec_vert_sys: Concrete: CIP
str_grid_long: 10.668
str_grid_short: 10.668
str_fdn_type: Deep foundation > 50' (15m)
str_sys_summary: Steel
lca_assessment_year: 2023
lca_design_phase: Construction
lca_rsp: 60
lca_software: Tally LCA
lca_purp_of_assessment: Voluntary: Internal Benchmarking
lca_phys_scope: BSECF
lca_ec_reductions: Yes
lca_ec_reduction_percent: 0.106
mass_total: 39350375.72180476
```

A.4 Task text v1 (baseline)

```
Scope: the building enclosure only, i.e. OmniClass "Shell - Enclosure" (exterior walls and cladding, exterior windows, curtain walls
      and storefronts, exterior doors, roofing, and their insulation). Exclude substructure, superstructure, interior construction
      and interior finishes. Life cycle stages: A1-A3 only (product stage: raw material supply, transport to the factory,
      manufacturing). Exclude A4, A5, B, C and D.

Provide a plausible range of the enclosure's A1-A3 embodied carbon in units of kgCO2e per m2 of building floor area (a range, not a
      single value). For floor area, use bldg_cfa. If bldg_cfa is not provided, use bldg_renovated_gfa + bldg_added_gfa. Briefly
      state your method and key assumptions.

End your answer with exactly one final line in this format and nothing after it:
RANGE_kgCO2e_per_m2: <low> - <high>
(plain numbers only, no units or other text on that line)
```

A.5 Task text v2

```
Scope. Estimate the embodied carbon of the building enclosure as classified in this dataset (OmniClass "Shell - Enclosure"), life
      cycle stages A1-A3 only (raw material supply, transport to the factory, manufacturing). Exclude A4, A5, B, C and D. The
      enclosure includes:
- exterior walls as complete assemblies: cladding, backup framing or studs, sheathing, cavity and continuous insulation, air and
      vapor barriers, and the interior gypsum layer of exterior walls; structural exterior walls (concrete, CMU, tilt-up) when they
      form the wall;
- fenestration: windows, curtain wall, storefront and skylights (frames and glass), and exterior doors;
- the roof assembly above the structural deck: insulation, cover board, membrane or roofing;
- insulation at slab edges, below grade and under slabs that belongs to the thermal envelope.
It excludes foundations, the structural frame, floor and roof decks or slabs, interior partitions and interior finishes.

Method. Work through these steps and show the key numbers:
1. Areas. If bldg_therm_env_area is given, use it as the total envelope area. Otherwise estimate the roof area (footprint = floor
      area / stories above grade), the above-grade wall area (perimeter x height) and the glazed area (bldg_wwr x wall area), plus
      any insulated ground-floor or below-grade area. State the resulting envelope-to-floor-area ratio; single-storey and low-rise
      buildings often exceed 1.
2. Assemblies. From the climate zone, R-values, building use, structure, region and construction type, choose the most likely wall,
      glazing and roof assemblies. Name the insulation type and estimate its thickness from the R-value.
3. Intensities. Give each component an A1-A3 intensity in kgCO2e per m2 of that component, using values typical of the North
      American whole-building LCA tools of the assessment year (Tally, One Click LCA). Account for these high-impact materials:
   - XPS insulation made with HFC blowing agents is common in these tools and carries roughly ten times or more the GWP of mineral
      wool, fiberglass or polyiso per unit of thermal resistance, so a thick XPS layer (for example roof or below-grade insulation in
      cold climates) can dominate the enclosure total.
   - Aluminum is carbon-intensive per kg (curtain wall and storefront framing, window frames, metal panels, ACM). Aluminum-framed
      curtain wall is about 100-200 kgCO2e per m2 of facade, and fully glazed curtain wall facades are among the highest-carbon
      enclosures.
   - Insulated glass units, precast concrete panels and masonry are also significant.
4. Total. Sum the components, divide by the floor area, and give a range that reflects your uncertainty about the assemblies. Do not
      pull the estimate toward a typical building: metal-clad, curtain-wall or thickly insulated enclosures can exceed 200 kgCO2e per
      m2 of floor area, and projects with little new enclosure work can fall below 10.
5. Renovations. For renovations and tenant improvements, count only new enclosure materials. Use bldg_proj_type, lca_phys_scope and
      mass_total per floor area to judge how much new envelope work was done. Major renovations often include new windows, cladding
      or roofing, and a high mass_total per floor area indicates substantial new material.
```

```
Floor area: use bldg_cfa. If bldg_cfa is not provided, use bldg_renovated_gfa + bldg_added_gfa.

End your answer with exactly two final lines in this format and nothing after them:
BEST_kgCO2e_per_m2: <value>
RANGE_kgCO2e_per_m2: <low> - <high>
(plain numbers only, no units or other text on these lines)
```

A.6 Task text v3 (final prompt)

Reference values are documented with sources, system boundaries and unit conversions in `references/envelope_intensities_sources.md`.

```
Scope. Estimate the embodied carbon of the building enclosure as classified in this dataset (OmniClass "Shell - Enclosure"), life
     cycle stages A1-A3 only (raw material supply, transport to the factory, manufacturing). Exclude A4, A5, B, C and D. The
     enclosure includes:
- exterior walls as complete assemblies: cladding, backup framing or studs, sheathing, cavity and continuous insulation, air and
     vapor barriers, and the interior gypsum layer of exterior walls; structural exterior walls (concrete, CMU, tilt-up) when they
     form the wall;
- fenestration: windows, curtain wall, storefront and skylights (frames and glass), and exterior doors;
- the roof assembly above the structural deck: insulation, cover board, membrane or roofing;
- insulation at slab edges, below grade and under slabs that belongs to the thermal envelope.
It excludes foundations, the structural frame, floor and roof decks or slabs, interior partitions and interior finishes.

Method. Work through these steps and show the key numbers:
1. Areas. If bldg_therm_env_area is given, use it as the total envelope area. Otherwise estimate the roof area (footprint = floor
     area / stories above grade), the above-grade wall area (perimeter x height) and the glazed area (bldg_wwr x wall area), plus
     any insulated ground-floor or below-grade area. State the resulting envelope-to-floor-area ratio; single-storey and low-rise
     buildings often exceed 1.
2. Assemblies. From the climate zone, R-values, building use, structure, region and construction type, choose the most likely wall,
     glazing and roof assemblies, and name the insulation type and thickness. Whole-building LCA models from this period (Tally, One
     Click LCA) often model roof and below-grade insulation as XPS made with HFC blowing agents. When the insulation type is not
     known, include that case in your range, especially in colder climates where the insulation is thick.
3. Intensities. Give each component an A1-A3 intensity in kgCO2e per m2 of that component. Use these reference values from EPDs and
     industry baselines (A1-A3, per m2 of component):
   Opaque walls (per m2 of wall, complete assembly):
   - steel stud + metal panel or ACM rainscreen: 70-110
   - steel stud + brick veneer: 90-120; brick veneer on CMU backup: 135-165
   - steel stud + EIFS: about 50; steel stud + fiber cement or stucco: 45-80
   - wood stud + vinyl or wood siding: 25-45; wood stud + brick or stone veneer: 60-100
   - insulated metal panel: newer products 27-72; legacy polyurethane (HFC) panels about 107 (2 in) to 274 (6 in)
   - precast concrete sandwich panel: 90-165; architectural precast with backup wall: 80-130
   - tilt-up concrete with interior insulation and gypsum: 80-125
   - curtain-wall spandrel: 75-140
   Fenestration (per m2 of unit, frame and glass):
   - aluminum stick curtain wall: 120-150; aluminum unitized curtain wall: 150-210
   - aluminum storefront: 140-155; aluminum window wall: 170-180; aluminum windows: 95-140
   - vinyl windows: 40-90; fiberglass windows: 75-145
   Roof above the deck (per m2 of roof, about R-25 to R-30, with membrane and cover board):
   - polyiso: 20-28; mineral wool board: 35-55; low-GWP XPS: 28-40; metal standing seam over polyiso: 27-40
   - XPS with HFC blowing agent: 85-110 when only product-stage emissions are counted, and up to 335-410 when the blowing-agent
     emissions are booked with the product, as some LCA tools do
   - insulated metal roof panel: 27-72 for newer products, about 190 for a legacy 4 in panel
   Insulation per m2 at RSI 1 (R-5.7): polyiso 2-4; EPS 2.5; mineral wool board 7-8; fiberglass batt about 1; closed-cell spray foam
     3-5 (HFO) or 20-36 (HFC); XPS 4-11 (low-GWP), about 18 (HFC, product stage only) or 58-99 (HFC, with blowing-agent emissions)
   Aluminum: about 6-13 kgCO2e per kg for North American extrusions; generic global aluminum data in LCA tools is about 16
Complete assemblies include framing, attachments, sheathing, barriers, sealants, flashing and trim, which early-stage take-offs tend
     to leave out; do not go below the low end of these ranges unless the assembly is clearly minimal.
4. Check. Convert your total to kgCO2e per m2 of envelope area and check that it is consistent with the reference values above. For
     new construction with conventional walls and 30-40% glazing, a blended value below about 50 kgCO2e per m2 of envelope usually
     means that a component, the aluminum and glazing content, or a high-carbon insulation option is missing; revisit those before
     you finalize.
5. Total. Divide the enclosure total by the floor area. Give a range that reflects your uncertainty about the assemblies, including
     the higher-carbon alternatives that are plausible for this building. Do not pull the estimate toward a typical building:
     metal-clad, curtain-wall or thickly insulated enclosures can exceed 200 kgCO2e per m2 of floor area, and projects with little
     new enclosure work can fall below 10.
6. Renovations. For renovations and tenant improvements the LCA covers only the new work, but that work often includes substantial
     new envelope elements (re-cladding, new curtain wall or windows, re-roofing). Do not assume the existing envelope is kept
     unless the data indicates it. Use bldg_proj_type, lca_phys_scope and mass_total per floor area as evidence of how much new
     material was installed.

Floor area: use bldg_cfa. If bldg_cfa is not provided, use bldg_renovated_gfa + bldg_added_gfa.

End your answer with exactly one final line in this format and nothing after it:
RANGE_kgCO2e_per_m2: <low> - <high>
(plain numbers only, no units or other text on that line)
```

A.7 Subagent instructions (template)

```
You are the language model being tested in a building life-cycle-assessment (LCA) experiment for a university course project. Your
      job is to answer a set of estimation prompts using only your own knowledge.

Input file: {task_file}
It is about {n_lines:,} lines, so read all of it with the Read tool, using offset/limit to page past line 2,000. It contains a
      System line, a shared context (a glossary of the feature names), and {n_buildings} building sections. Each section starts with
      a line like `===== KEY: {example_key} =====`, followed by the building's features and the question.

Rules. Breaking any of these invalidates the experiment:
- Read only that one task file. Do not open, list or search any other file or folder (especially nothing under results/,
      GWP-calculation/ or references/). Do not use web search or web fetch, and do not run code or shell commands. The only tools you
      may use are Read (on the task file) and Write (for your answer files).
- Treat each building as a separate, independent request, as if you were seeing it alone. Do not compare buildings, rank them, or
      adjust one estimate because of another.
- Follow each building's prompt exactly: {answer_length}, then end with {final_lines} containing plain numbers only.

Output: write your answers in parts of about 10 buildings each, using one new file per part and never overwriting an earlier part:
{answers_prefix}_part-1.md, then _part-2.md, and so on.
Inside each part file, every answer starts with its key line copied exactly from the task file, followed by the answer text, for
      example:

===== KEY: {example_key} =====
<answer>
{example_final}

Answer all {n_buildings} buildings, in the order they appear in the task file, with the key lines copied exactly.

When you are finished, reply with exactly one line: `done: <number of answers written> answers in <number of parts> parts`. Do not
      include the answers themselves in your reply.
```

B Reproduction

```
python GWP-calculation/sum_gwp.py                  # ground truth (enclosure, A1-A3)
python GWP-calculation/enclosure_breakdown.py      # material breakdown + data-quality flags
python subagent_runs.py prepare --run-name 20261104_prompt-v3 --runs 3 --effort medium
#   one Claude Code subagent (Opus, effort medium) per results/<run>/subagent/tasks/*.instructions.txt
python subagent_runs.py resume  --run-name 20261104_prompt-v3   # only if interrupted
python subagent_runs.py collect --run-name 20261104_prompt-v3
python analysis/make_split.py
python analysis/compare_runs.py 20261104_baseline-innate 20261104_prompt-v2 20261104_prompt-v3
python analysis/error_analysis.py --split dev 20261104_baseline-innate 20261104_prompt-v2 20261104_prompt-v3
python analysis/feature_study.py
python report/make_figures.py
```